Probabilistic Inference with Algebraic Constraints

Theoretical Limits and Practical Approximations

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*December - **NeurIPS 2020** Spotlight*

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Motivation

Example: Skill matching system

Minka et al., "Trueskill 2: An improved bayesian skill rating system", 2018

Motivation

Example: Skill matching system

- Each ***player*** has a certain skill
  $\Rightarrow$ *continuous variables*

Minka et al., "Trueskill 2: An improved bayesian skill rating system", 2018

Motivation

Example: Skill matching system

$0 \leq X_{P_i} \leq 10$
for $i = 1, \ldots, N$

Minka et al., "Trueskill 2: An improved bayesian skill rating system", 2018

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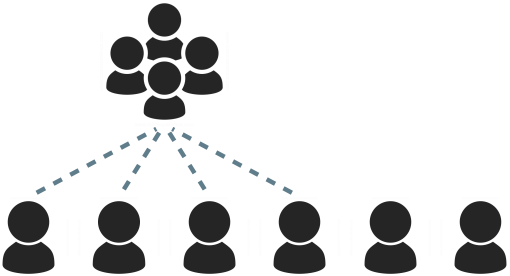
Example: Skill matching system

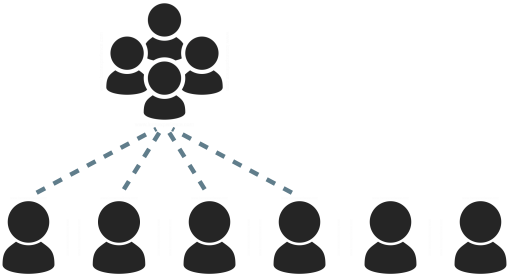


- $0 \leq X_{P_i} \leq 10$
  for $i = 1, \ldots, N$
- Players can form ***teams***
  $\Longrightarrow$ *complex constraints*

Minka et al., "Trueskill 2: An improved bayesian skill rating system", 2018

Motivation

Example: Skill matching system



- $0 \leq X_{P_i} \leq 10$
  for $i = 1, \ldots, N$
- $\mid X_{T_j} - X_{P_i} \mid < 1$
  for $j = 1, \ldots, M, i = 1, \ldots, |T_j|$

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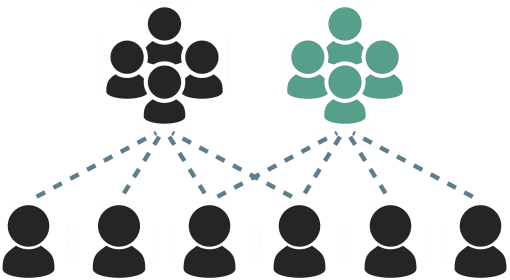
Motivation

Example: Skill matching system

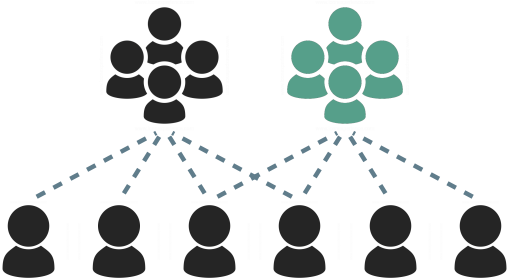


- $0 \leq X_{P_i} \leq 10$
  for $i = 1, \ldots, N$
- $\mid X_{T_j} - X_{P_i} \mid < 1$
  for $j = 1, \ldots, M, i = 1, \ldots, |T_j|$
- Good teams form a ***squad***
  $\Longrightarrow$ *discrete variables*

Minka et al., "Trueskill 2: An improved bayesian skill rating system", 2018

Motivation

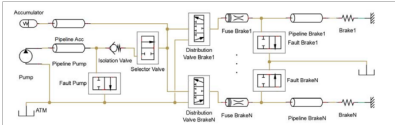
Example: Skill matching system

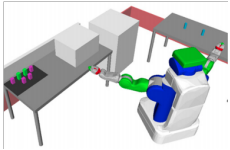


- $0 \leq X_{P_i} \leq 10$
  for $i = 1, \ldots, N$
- $\mid X_{T_j} - X_{P_i} \mid < 1$
  for $j = 1, \ldots, M, i = 1, \ldots, |T_j|$
- $B_{S_j} \Rightarrow X_{T_j} > 2$
  for $j = 1, \ldots, M, i = 1$

Minka et al., "Trueskill 2: An improved bayesian skill rating system", 2018

Continuous + discrete + constraints = SMT





Satisfiability Modulo Theories of linear arithmetic over the reals (SMT($\mathcal{LRA}$)) delivers all the ingredients by design!

Widely used as a representation language for ***robotics***, ***verification*** and ***planning*** *[Barrett et al. 2010]*

Barrett et al., "Satisfiability modulo theories", 2018

Continuous + ***discrete*** + ***constraints*** = **?**

Continuous + ***discrete*** + ***constraints*** = ?

Generative adversarial networks (GANs) *[Goodfellow et al. 2014]*
Variational Autoencoders (VAEs) *[Kingma et al. 2013]*

Hybrid Bayesian Netowrks (HBNs) *[Heckerman et al. 1995; Shenoy et al. 2011]*
Mixed Probabilistic Graphical Models (MPGMs) *[Yang et al. 2014]*

Tractable Probabilistic Circuits (PCs) *[Molina et al. 2018; Vergari et al. 2019]*

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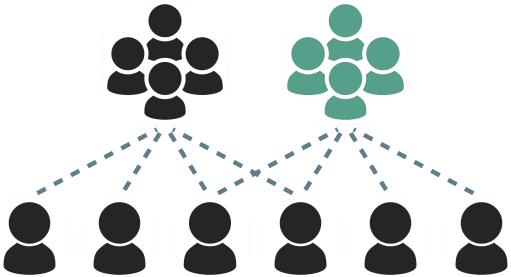
Continuous + **discrete** + **constraints** = **SMT**



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Barrett et al., "Satisfiability modulo theories", 2018

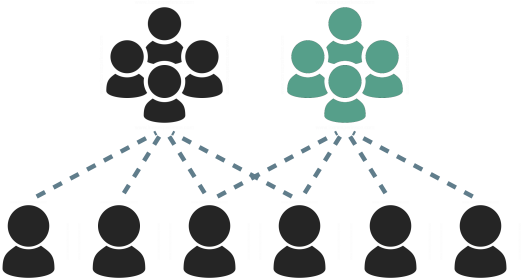
Continuous + **discrete** + **constraints** = **SMT**

$$\Delta = \bigwedge_i 0 \leq X_{P_i} \leq 10 \bigwedge_j \bigwedge_{i \in T_j} \mid X_{T_j} - X_{P_i} \mid < 1 \bigwedge_j (B_{S_j} \Rightarrow X_{T_j} > 2)$$

a single CNF SMT($\mathcal{LRA}$) formula Δ...

Barrett et al., "Satisfiability modulo theories", 2018

Continuous + **discrete** + **constraints** = **SMT**



"What is the probability of team T_1 outperforming team T_2, if T_1 is a squad but T_2 is not?"

SMT + weights

$$\bigwedge_i 0 \leq X_{P_i} \leq 10$$

$$\bigwedge_j \bigwedge_{i \in T_j} \mid X_{T_j} - X_{P_i} \mid < 1$$

$$\bigwedge_j (B_{S_j} \Rightarrow X_{T_j} > 2)$$

+

$$\begin{cases} w(X_{P_i}), \\ \quad \text{if } 0 \leq X_{P_i} \leq 10 \\ \\ w(X_{T_j}, X_{P_i}), \\ \quad \text{if } \mid X_{T_j} - X_{P_i} \mid < 1 \\ \\ w(B_{S_j}, X_{T_j}), \\ \quad \text{if } B_{S_j} \Rightarrow X_{T_j} > 2 \end{cases}$$

SMT formula Δ

weight functions $\mathcal{W}$

Belle et al., "Probabilistic inference in hybrid domains by weighted model integration", 2015

SMT + weights = Weighted Model Integration

$$\bigwedge_i 0 \leq X_{P_i} \leq 10$$

$$\bigwedge_j \bigwedge_{i \in T_j} \mid X_{T_j} - X_{P_i} \mid < 1$$

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complex support

+

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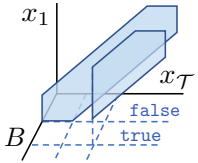
densities

=



(unnormalized)

$\mathsf{Pr}_{\Delta}(\mathbf{X}, \mathbf{B})$

Belle et al., "Probabilistic inference in hybrid domains by weighted model integration", 2015

SMT + densities = Weighted Model Integration

Given an SMT($\mathcal{LRA}$) formula Δ over continuous vars $\mathbf{X}$ and discrete ones $\mathbf{B}$, and weight function $\mathcal{W}$, the ***weighted model integral*** (WMI) is

$$\mathsf{WMI}(\Delta, \mathcal{W}; \mathbf{X}, \mathbf{B}) \triangleq \sum_{\boldsymbol{b} \in \mathbb{B}^{|\mathbf{B}|}} \int_{(\mathbf{x}, \boldsymbol{b}) \models \Delta} w(\mathbf{x}, \boldsymbol{b})\, d\mathbf{x}.$$

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$\Longrightarrow$ *integrating the* ***densities*** *of the* ***feasible regions*** *of* Δ*!*

i.e., computing the ***partition function*** of the unnormalized distribution $\Pr_\Delta$

Belle et al., "Probabilistic inference in hybrid domains by weighted model integration", 2015

"What is the probability of team T_1 outperforming team T_2, if T_1 is a squad but T_2 is not?"

Belle et al., "Probabilistic inference in hybrid domains by weighted model integration", 2015

WMI

Advanced probabilistic reasoning

$$\Phi_S : (B_{S_1} = 1 \wedge B_{S_2} = 0) \implies T_1 \text{ is a squad, } T_2 \text{ is not}$$

$$\Phi_T : (X_{T_1} > X_{T_2}) \implies T_1 \text{ outperforms } T_2$$

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WMI

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$$\Phi_T : (X_{T_1} > X_{T_2}) \implies T_1 \text{ outperforms } T_2$$

$$\mathsf{Pr}_\Delta(\Phi_T \mid \Phi_S) = \frac{\mathsf{WMI}(\Delta \wedge \Phi_T \wedge \Phi_S, \mathcal{W})}{\mathsf{WMI}(\Delta \wedge \Phi_S, \mathcal{W})} = \frac{4,206}{7,225} \approx 58.22\%$$

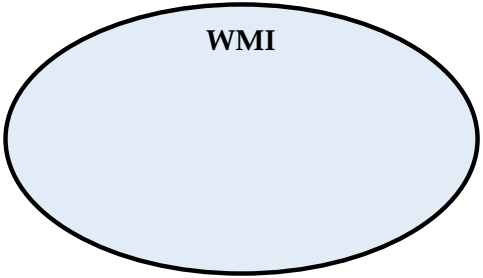
$\Longrightarrow$ *conditional probabilities as a ratio of two weighted model integrals*

Belle et al., "Probabilistic inference in hybrid domains by weighted model integration", 2015

Tractability of WMI

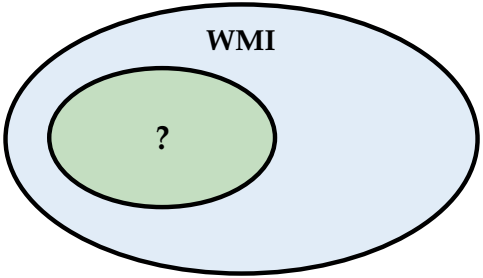
Why is building inference algorithms for hybrid domains difficult?

Tractability of WMI



- ***#P-hard*** in general

Tractability of WMI



#P-hard in general

what would be tractable?

Primal Graph

Discrete Graphical Models

$$\bigwedge_{i=1,2} (X_i \Rightarrow X_{i+1})$$

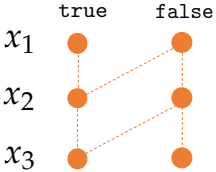
WMI models

$$\bigwedge_{i=1,2} \{(X_i - 0.1 \leq X_{i+1} \leq X_i + 0.1) \vee (X_i + 0.9 \leq X_{i+1} \leq X_i + 1.1)\}$$

Primal Graph

Tree Primal Graph

Discrete Graphical Models



tractability ✔

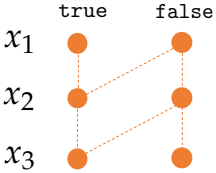
WMI models

?

Primal Graph

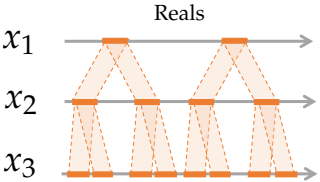
Tree Primal Graph

Discrete Graphical Models



tractability ✔

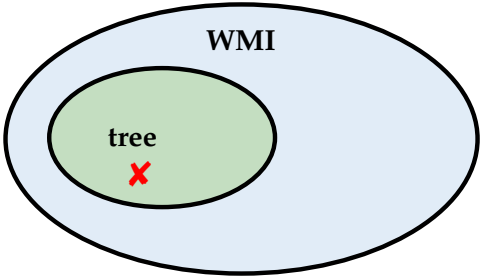
WMI models



tractability ✘

Primal Graph

Tractability of WMI

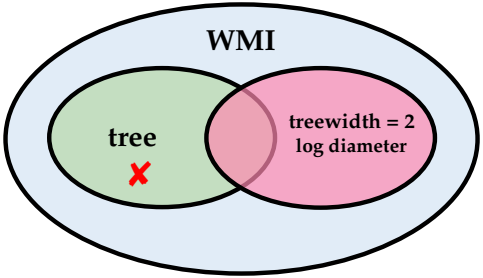


#P-hard in general

tree WMI problem class ✘

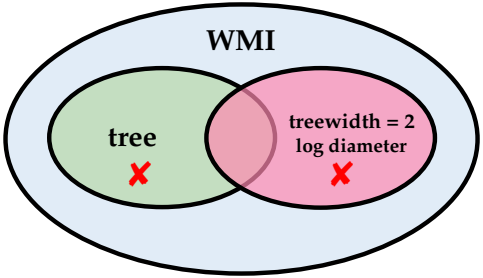
WMI Inference on tree-shaped primal graphs with unbounded-diameter is #P-hard!

Tractability of WMI



- ***#P-hard*** in general
- tree WMI problem class ✘
- logarithmic diameter and treewidth two?

Tractability of WMI



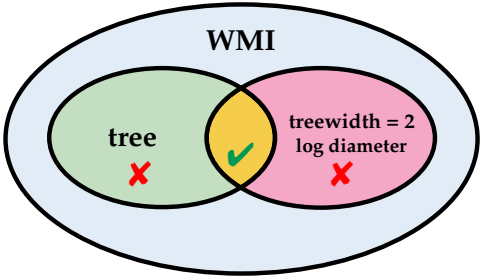
#P-hard in general

tree WMI problem class ✘

logarithmic diameter and treewidth two ✘

WMI inference on primal graphs with bounded-diameter but treewidth two is #P-hard!

Tractability of WMI



#P-hard in general

tree WMI problem class ✘

logarithmic diameter and treewidth two ✘

intersection ✔ *[Zeng et al. 2020]*

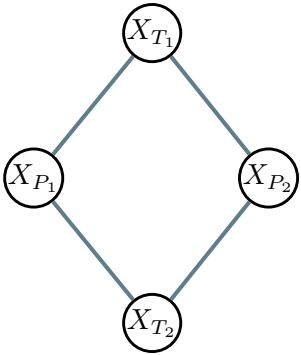
...but how can we perform inference on general WMI problems?

ReCoIn

Approximate WMI Inference

ReCoIn

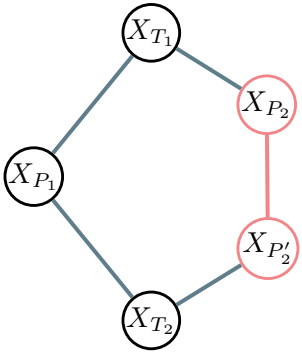
Approximate WMI Inference



- Given a WMI problem with ***loopy primal graph***

ReCoIn

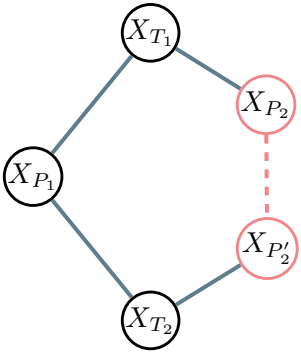
Approximate WMI Inference



- Given a WMI problem with ***loopy primal graph***
- ***Re***lax it by adding ***copies*** of literals

ReCoIn

Approximate WMI Inference

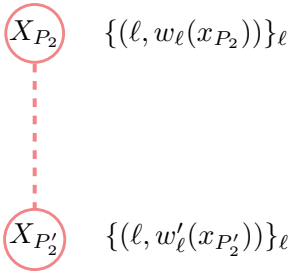


- Given a WMI problem with ***loopy primal graph***
- ***Re***lax it by adding ***copies*** of literals, then removing equality constraints

  $\Longrightarrow$ *removing dependencies, breaking loops*

ReCoIn

Approximate WMI Inference



- Given a WMI problem with ***loopy primal graph***
- ***Re***lax it by adding ***copies*** of literals, then removing equality constraints
- ***Co***mpensate for the removed dependencies, by introducing certain literals and weights

ReCoIn

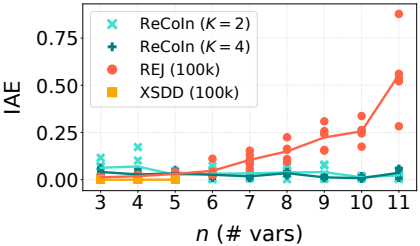
Approximate WMI Inference

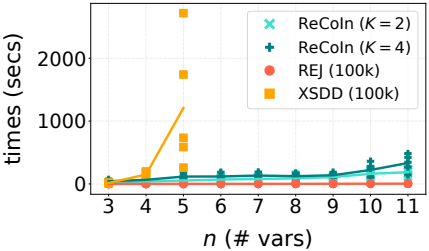
$$w_\ell \leftarrow f(\mathsf{Pr}_\Delta(\ell); w')$$

$$w'_\ell \leftarrow f(\mathsf{Pr}_\Delta(\ell); w)$$

- Given a WMI problem with ***loopy primal graph***
- ***Re***lax it by adding ***copies*** of literals, then removing equality constraints
- ***Co***mpensate for the removed dependencies, by introducing certain literals and weights
- optimize compensating weights iteratively by solving a series of ***exact In***tegration problems

Experiments





$\Longrightarrow$ ReCoIn ***scales better*** *to larger WMI problems while still* ***delivering accurate approximations***

Conclusions

Real-world data is ***noisy***...

Conclusions

Real-world data is ***noisy***, ***complex***...

Conclusions

Real-world data is ***noisy***, ***complex*** and ***mixed continuous-discrete***...

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The WMI framework is very appealing for probabilistic inference in the real-world!

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Efficient approximations are not only useful, but ***needed***

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$\Longrightarrow$ *ReCoIn delivers fast approximate inference*

Conclusions

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Next

Application to program verification, probabilistic (logic) programming, ...

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Efficient approximations are not only useful, but ***needed***

$\Longrightarrow$ *ReCoIn delivers fast approximate inference*

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Application to program verification, probabilistic (logic) programming, ...

Code

`github.com/UCLA-StarAI/recoin`

References I

- Heckerman, David and Dan Geiger (1995). “Learning Bayesian networks: a unification for discrete and Gaussian domains”. In: *Proceedings of the Eleventh conference on Uncertainty in artificial intelligence*. Morgan Kaufmann Publishers Inc., pp. 274–284.
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